

FLIGHT, HISTORY OF

EVERY DAY, MILLIONS OF PEOPLE fly to destinations all over the world. Planes are a common sight in the skies, but, despite their widespread use, they were first developed only about 90 years ago. The urge to fly is ancient, but by the start of the 20th century, the only flying machines were hot-air balloons, airships, and gliders. World War I stimulated the development of aeroplanes and, by the end of World War II, advances had resulted in jets and rockets. Since then, flight technology has produced supersonic planes and space travel.

First controlled flight

The first controlled flight of a powered aeroplane took place on 17 December 1903 in Kitty Hawk, North Carolina, USA. The plane, *Flyer I*, flew 36 m (119 ft) in a flight that lasted under 12 seconds: it is nonetheless perhaps the most famous flight of all time. *Flyer I* was designed and built by the Wright brothers, Orville and Wilbur, after years of experiments with kites and gliders. It was powered by a petrol engine they built themselves.



Flyer I climbed to a height of 3 m (10 feet).

Wilbur Wright watches his brother Orville take off.

Warplanes

The military's interest in the potential of aeroplanes as weaponry was central to the advancement of flight technology. During World War I (1914–18), warplanes were transformed from being slow and vulnerable to being fast, easily manoeuvrable fighting machines. Huge bomber and fighter planes were made in this period.



Protective clothing worn by World War I pilots included flying helmet, goggles and gauntlets.



Biplane of 1917

Biplane

World War I pilots flew biplane (twin-winged) fighters. Built from wood and fabric, biplanes were sturdier than monoplanes, but flying them was little fun. The cockpit was open to the cold and wet, and to spits of oil from the engine. Larger fighters had a second cockpit for a navigator and gunner. Single-seaters had a machine gun that fired through the spinning propeller.



Airships

Airships are held aloft by a vast gas-filled envelope and driven forward by engines with propellers. Airships were an important form of passenger and military transport until long-distance aircraft were developed in the 1940s.

Modern age

The basis for the modern aeroplane first appeared in the 1920s. It was a monoplane (single-winged) aircraft. The wing was made of metal, as was the fuselage. All aircraft had piston engines and propellers until the late 1930s, when a new type of engine, the jet, was invented.



Boarding pass

Airline ticket



Harrier GR5 jet fighter

Jets

A turbojet engine allows aircraft to fly much faster and more quietly than a propeller engine. Jet aircraft were increasingly used after World War II and became standard for fighter aircraft and for long-distance passenger planes.

Passenger flight

Fast, comfortable, and affordable air travel had become accessible by the 1960s. Today, millions of passengers fly around the world in the Boeing 747, a so-called "jumbo jet" which has quiet, turbofan engines.

Space flight

The first rockets powerful enough to reach space were built in the late 1950s. Today, modern launch vehicles and re-usable spacecraft, such as the Shuttle, make going into orbit almost an everyday event.

Ariane launch vehicle



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